3D-VISUALIZATIONS AS A MEANS FOR ENGAGING USERS AND ACTORS AS CO-DESIGNERS IN THE FUZZY FRONT-END OF PRODUCT DEVELOPMENT

Snorre Hjelseth,
The Oslo School of Architecture and Design, Vestfold University College, Norway.
snorre.hjelseth@hive.no

ABSTRACT

This paper explores the role of photorealistic images created by computer generated 3D models when designing complex systems in the fuzzy front-end of innovation. It discusses how realistic visualizations can engage users and actors as co-designers in early stages of product development. The paper introduces the SimSam project as a case where participation from expert-users and stakeholders has been critical for the decisions made in the design process and the ability to implement the selected design. The computer generated visualizations were a key element in engaging these participants in the front-end of the process. This paper takes up this position with reference to computer generated 3D representations and their potential in helping realize complex relations between tools, participants and representations in product development. The results show that the 3D-visualizations had an impact on involving participants in the design tasks and when implementing the concept on internal and external levels.

Keywords: Photorealistic representations, co-design, product development.

INTRODUCTION

This paper reports results of an exploratory case study where photorealistic 3D computer generated images were used as a means for engaging users and actors as participants in a co-design project in the Fuzzy Front End (FFE) of product development (PD). Below I describe how the participants were able to contribute to the design process and how this result affected the implementation of the design itself. The central question the paper poses is: How can photorealistic images created with computer generated 3D models be used as a means to engage users and actors as co-designers in the front end of PD?

The paper refers to a case called SimSam that covers a design process where a new type of innovation and simulation-lab was designed. SimSam was devised to explore and to develop possible new solutions for maritime innovation processes. In order to develop SimSam, a multidisciplinary team of designers, engineers and users was established. Photorealistic images of the possible designs and product scenarios were produced using computer generated 3D models in order to facilitate communication and collaboration between the participants.

DESIGN IMPACT IN THE FRONT END OF INNOVATION

Innovation might be seen as a combination of two factors. One is the invention part, where previous ideas are combined into new ones. The second part is the ability of implementing ideas into the world and is usually seen as the economical part of the innovation process. The innovation processes in PD might be seen as having three phases: the fuzzy front end, new product development and the commercialization phase (Koen, 2004). It is in the FFE of innovation it is most likely to create breakthrough products from both of the perspectives of invention and implementation. A breakthrough product might be defined as a product that is new to the world and that is several cycles further in development than the output of an incremental process. In the FFE it is possible to respond to costumers needs, wants and desires with most impact (Cagan and Vogel, 2001). A company environment allowing employees to be open and speak their mind and be trusted are important

factors to stimulate breakthrough discoveries in the FFE (Koen, 2004).

The design space in the front end of innovation is complex. Often problems are ill-defined or wicked and have a number of unknown factors. The designers` way of dealing with this type of problem is described as solution driven design (Lawson, 2005; Lawson and Dorst, 2009). Through creation of solutions to ill-defined problems it is possible to reflect back on the factors that define the problems.

DESIGN AS A COLLECTIVE PROCESS

Often when working with breakthrough solutions in the FFE, the process itself becomes complex. Some of this complexity deals with the need to integrate multidisciplinary knowledge in a design process. Tim Brown explains it like this: "a competent designer can always improve on last years new widget, but an interdisciplinary team of skilled design thinkers is in a possession to tackle more complex problems". (Brown, 2009: 7)

In design research the collective and collaborative process in a multidisciplinary team is referred to as Co-design: "Co-design is the process in which actors from different disciplines share their knowledge about both the design process and the design content. They do that in order to create shared understanding on both aspects, to be able to integrate and explore their knowledge and to achieve the larger common objective: the new product to be designed". (Kleinsmann, 2006: 30)

When design becomes multidisciplinary the collaboration and communication within the development team becomes intricate, and a critical part of the process is to obtain a shared understanding between design participants (Bucciarelli, 1994). The knowledge between participants is shared orally with the support of representations like drawing and models (Kleinsmann and Valkenburg, 2008).

In contrast to the user-centered design tradition, co-design is about integrating all members that has a stake in the product to be involved in the design process. This means that users are not only observed, but are active participants in solving design problems. A user who is involved in the design task might be seen as an expert on his or her area of expertise. This expertise becomes an important part of knowledge development, idea generation and concept development. In such co-design contexts the designer gets the role of a facilitator that generates tools for ideation and expression (Sanders and Stappers, 2008). This process might also be referred to as collective creativity (Sanders and Rim, 2001). To create shared understanding in a co-design process involving users is more complex than within a group of experienced developers. In contrast to designers, users have little experience in development processes where representations play a critical role. Representations created in the FFE are often abstract and are materialized through sketches, drawings, mock-ups or physical models. If the co-design process is going to become successful it is critical that the user understands these representations.

Representations are used in product design processes in order to share ideas and thoughts. In the user centered design tradition the user is often studied as an object and not engaged in the design process itself. Participatory design, co-design, co-creations and user driven innovation are areas in design research that explore such collaboration.

REPRESENTATION TYPES

Different types of representations used in PD have different qualities. These representations can be 2D drawings and 3D models, both physical and virtual. Representations have common factors that influence the way they are perceived, that can be described as criteria for assessment of visual representations. Table 1 shows the criteria proposed by different authors (Bates-Brkljac, 2010).

Such criteria are scalable. For example, a sketch can be very accurate or be imprecise. The criteria can also be mixed. Another example; a rendering might look very realistic, but might lack accuracy in dimensions. Visual representations might be categorized in many different ways. Categorizations might be grouped in professions, medium or subject (Buxton, 2007). Buxton describes five types of

renderings: Sketch, memory drawing, presentation drawing, technical drawing and description drawing. These rendering types can also refer to physical and virtual 3D models.

3D software is usually not designed for design tasks in the creative ideation phases. Representations may have the characteristics of finished products, which might limit the design space for new creative solutions (Parsons, 2009). Detailed representations might be perceived as prototypes and draw attention to details that disturb the creative process and possible opportunities (Buxton, 2007).

The levels of accuracy and realism of representations created in the front-end might become a paradox in a collective design setting. If the representations are abstract, participants might lack the skills to read the representations (Powell, 1994). On the other hand, if the representations have a high level of accuracy and realism it might limit the creative design space.

VISUAL SIMULATION

Creating realistic images to represent ideas in a design process can be described as a visual simulation. Al-Kodmany (1999) has conducted research into the effects of using photorealistic representations created by computer photo-manipulations to engage public participation in urban planning. The results from this study show that the photorealistic "visualizations created through digital technology provided a common language for all participants", this resulted in a rise of excitement from the users and a design that "reflected the community's wishes and input and respected their cultural heritage". (Al-Kodmany, 1999: 45) In a similar way, the research conducted in the SimSam case investigates how the use of photorealistic visualizations engages users and actors as active participants in a design process.

RESEARCH METHODS

During the design process of SimSam the author had the role of both a researcher and a designer. This way of actively involving the researcher in the activity that is being researched is referred to as participatory action research (PAR) (Hult and Lennung, 1980; Denzin and Lincoln, 2000).

The empirical data is produced through case studies in real-life situations (Yin, 2009). The participants in the case study were interviewed using a qualitative interview method (Kvale and Rygge, 2009). The interviews focused on the roles of the different actors in the design process, the collective design process and the use of representations. The interviews were recorded using a video camera and transcribed. The examples from the transcriptions have been translated from Norwegian to English by the author. Four actors were interviewed. The actors had the roles of a user, a designer, a project leader and an external stakeholder. They all were actively involved in the design process, where the external stakeholder had the role of a possible investor.

The research is based on an exploratory case study. This approach was necessary in order to explore different research angles when researching the subject, and at the same time work as a designer with the goal of making the SimSam-lab a success.

The criteria for the assessment of visual representations proposed by different authors

Appleyard	Sheppard	Pietsch	Radford
Realism	Accuracy	Abstraction	Abstraction
Accuracy	Representativeness	Accuracy	Accuracy
Comprehensibility	Visual clarity	Realism	Realism
Evaluability	Interest		
Engagement	Legitimacy		

Table 1. The criteria for assessment for visual representations proposed by different authors. Author reference, (Appleyard, 1977; Sheppard, 1989; Radford, 1997; Pietsch, 2000; Sheppard, 2001).

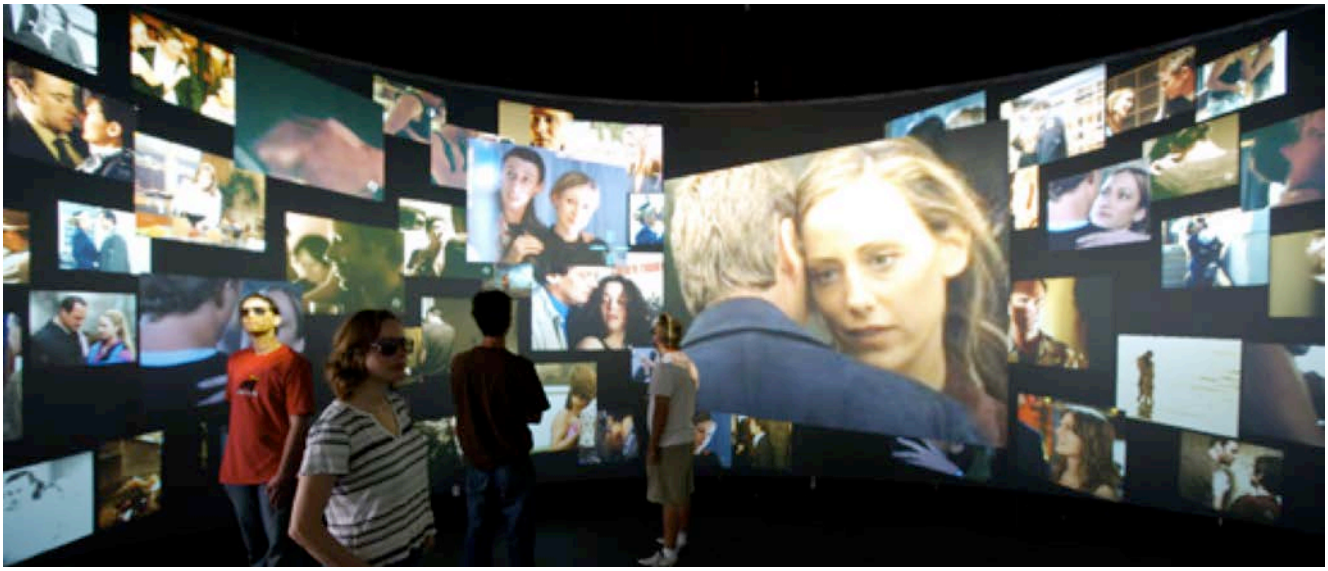


Figure 1. T_Visionarium in iCinema by Dennis Del Favero, Jeffrey Shaw, Neil Brown, Peter Weibel, Matt McGinity 2008. Video link: http://vimeo.com/28322411

THE CASE

BACKGROUND

The SimSam project started out as ideas about a future innovation facility. The concept was generated in parallel with Vestfold University College`s (HiVe) development of a new research and innovation center named FIN. This new R&I center was to house a new ship simulator as part of the new research and simulator development strategy in the maritime sector. A ship simulator is basically a large multi-projected cylindrical curved screen that displays the output of a virtual 3D-environment that is interacting with physical hardware consoles, which are standardized on ships. In the early summer of 2009 SimSam was suggested as an idea where the new ship simulator could be combined with design methods and thinking. This new idea about an innovation-lab was named SimSam and stands for simulation and collaboration (in Norwegian).

Inspirations for the idea came from The Oslo School of Architecture and Design, which had formed a similar concept named C-Lab. The ideas behind this lab were based on the use of design methods and thinking in combination with state-of-the-art visualization and modeling technology.

DESIGN METHODS AND TECHNOLOGY

In accordance with Capjon (2004) the design methods and thinking used in SimSam builds upon the use of representations as stimuli for expression and collaboration in PD processes. In order to reflect on and communicate ideas co-actors need to embody representational experiences through perceptual stimulation of the human senses. This involves that internal ideas must be externalized through a medium. Most often externalizations are combined in order to explain ideas meaningfully. The media in which the representations are created are critical and need to be adjusted according to the information being communicated. A representation that involves too little or too much complexity might change the understanding of an idea completely. Such representations might range from verbal communications to sketches and functional prototypes. If such representations can be easily modified on-the-fly, it is possible to obtain a dynamic creative interaction between participants. Industrial designers have special skills and knowledge in creating such types of representations.

Creativity can be understood as combining previous ideas into new ones. When such processes are taking place it can be a beneficial to make representations of these ideas and to make them available for all the participants (Figure 1). By being able to visualize multiple ideas and concepts simultaneously, they can form a platform for comparing and combining concepts into completely new ideas.

The concept of solution-driven design processes is one of the core principles built into SimSam –

through offering tools that enable users to create fast models and reflect back on problems in the design space. Much of the information implemented into PD consists in a digital format. The screens and computers that are commonly used allow only a limited amount of resolution on a limited space. The limited resolution allows you to display a limited amount of information. This is because most digital information and media are designed for computers used by individuals on single screens. If one uses a larger screen like a projector the resolution is often limited to 1080p HD.

By combining multiple projectors it is possible to create a large screen with a higher resolution. This enables the possibility of visualizing large amounts of information and media simultaneously. Figure 1 shows the iCinema project from the University of New South Wales in Sydney. The iCinema has the capability of playing hundreds of videos simultaneously on the same screen.

The interaction with the multi projection screen is a critical element involving user participation in SimSam. A traditional solution to such interaction is through a mouse and keyboard interface. The problem with the traditional interface is that it is restricted to single-user interaction. The SimSam philosophy involves that all participants in an appropriate PD process should be able to use and interact with the representations. A user interface that is more open for multiple user interaction might increase the level of user participation and might help the setting to feel less static and over-controlled by a facilitator. A touch-sensible tabletop was a solution we found for this problem. If we used a 50-inch screen we found that it enables multiple users to share the space around the interface with multiple interaction inputs. One capability of this type of interface is that it allows digital representations to be at the same level of accessibility for the whole development team.
Up to this point the SimSam was only an invention of how to combine new technologies in new ways. The context in which it should be used was unclear and there had been no decision about taking the idea further. To take the idea to the next level in the development process we needed to se how it could be implemented in real life scenarios.

FACILITATING A COLLECTIVE DEVELOPMENT PROCESS

If the SimSam was going to be implemented it needed to share facilities with a ship simulator. To implement the idea we needed to see if it was possible to combine the SimSam idea with some existing and future activities at HiVe. To do this we involved colleagues from different maritime departments to join the process and see if there were any possible scenarios to be implement into the SimSam concept. To communicate these ideas about how to use this technology for PD we needed some representations that explained the concept with more fidelity than an abstract sketch – because involved actors had to be able to understand what the technology looked like and the dimensions of the facility itself.

Using a traditional representation tool in this phase like hand sketches, physical scale models or mock-ups was problematic. The complexity of the technology and the dimensions of the SimSam challenged the representation principle. One of the representations needed to capture the overview of the technology put together in a room, and some of the representations needed to be on more detailed level of interaction between participants using the



Figure 2. This picture shows the SimSam-lab idea before the scenarios was developed and how the different technologies might be put together. By using the layered structure of the 3D models it was possible to make rapid changes to the visualizations.

technology (Figure 2). We also needed representations that were possible to change rapidly in order to visualize different scenarios.

We tried to involve actors from the maritime field. These actors had little experience in PD and using abstract sketches or drawings. The goal was to get a discussion going about the implementation of maritime applications, but we were not able to obtain a shared understanding of the SimSam idea. By creating a virtual 3D-model of the SimSam idea we were able to get a more detailed view over the technology assembled. We also placed models of people (personas) in the room to get a feeling of the dimensions and to create scenarios of interaction. After constructing the 3D-models in Catia and assembled them in Maya, we textured the models and rendered images from different camera views. The pictures were presented to the SimSam development team through printouts and monitors.

After presenting images of the 3D-model to the SimSam development actors, they became more enthusiastic about the idea. It was easier to get a shared understanding of the different technologies and how we were thinking about using it. After some discussion we started to form some interesting scenarios of how the SimSam-lab might be used. We created one oil-spill scenario where the lab can be used to simulate an oil-spill (Figure 3). The visualized scenarios show how multi projection screens and the tabletop can be utilized. The multi projection screen were able to show large amounts of data simultaneously that are critical for the operation and the tabletop made it possible for the actors to interact and simulate boats trying to collect and clean up the oil. The oil-spill scenario was engaging because of two recent oil-spills near the Norwegian coastline. The handlings of the cleaning operations have been heavily criticized because of deficient organizing skills.

Another visualized scenario was a ship design process of environment friendly ships (Figure 4). With this scenario we wanted to visualize how the SimSam lab is able to handle large amounts of information and how different design concepts can be compared and become part of a multidisciplinary collaboration.

One important aspect of this way of facilitating the design process was to combine the use of ship simulation technology to test and review the different designs.

IMPLEMENTING THE DESIGN

From ideas about design thinking and methods in combination with state-of-the-art technology the SimSam-lab evolved into concepts involving context and action scenarios. These scenarios were presented to the HiVe board, which in response decided to implement the concept as part of the new FIN R&I center. They decided to expand the project and introduced the project to the MarKIS network.

The MarKIS network is a maritime network combining commercial companies, universities and institutions from Denmark, Norway and Sweden. The purpose of the network is to establish a maritime collaboration network across Scandinavia. The SimSam concept was introduced to the MarKIS as a possible new way of working with maritime development. HiVe and MarKIS made a decision to collaborate with the development of SimSam and funded the project with 1.3 million euros.



Figure 3, The picture show an oil-Spill scenario simulation where information is placed on the multi-projection wall and a tabletop is used to combine physical representation with digital representations.



Figure 4. Pictures of eco ship design scenario.

RESULTS

The results from the conducted research where analyzed and it shows that the 3D visualizations had an impact on three different levels in the FFE of the SimSam development process. The different levels can be described as *internal* (the SimSam development team), *company level* and on an *external level* with external companies and organizations.

LEVEL ONE: THE DESIGN TEAM

Level one is the use of 3D visualizations to support communication and collaboration within the multidisciplinary development team. The issue regarding visualizations and realism in the FFE is that hi-fidelity in realism might have a negative impact on the creativity space. Based on PAR observations and the analyzed interviews, it was found that the 3D computer generated representations with a high level of realism impacted on the design process and the implementation process of the design concept. The results show that the design team actors were able to create a shared understanding using the 3D representations and involve users as contributors to the design.

When the designer was interviewed and asked how the different representations were understood, he answered: “External partners and users were not able to understand the 2D sketches, but this changed when we started using the 3D visualizations”. The designer also described that there was a change in enthusiasm on the part of the users when we started using the photorealistic 3D representations: “They were open to the concept, but their enthusiasm was limited because they did not see the possibilities in the maritime sector, but when we started to discuss the simulator using the 3D visualizations their understanding of the concept started to grow. They started then to see new possibilities using the concept within their maritime field”. This statement was later confirmed in the interview with the user.

When the project proceeded from the FFE to the new PD phase, new modifications were made to the design. These changes were represented with a 2D plan drawing and not as the earlier photorealistic images. The user representative stated through the interview that he had problems relating to the plan drawing and he had problems relating his knowledge to the new design.

Going from abstract visualizations to realistic representations and back again to abstract visualizations leaves some evidence on how the users were able to interact and participate during the design process. The enthusiasm the user experienced from interacting with the realistic visualizations might be due to the seductive visual qualities and the ability to realize their own thoughts and ideas simulated through realistic visualizations. At the same time, the participants were able to reach a shared understanding of the SimSam ideas, and to create scenarios of possible maritime applications together, based on real-life experiences. The development of these scenarios would not have been possible without the involvement of users from the maritime sector.

LEVEL TWO: THE COMPANY LEVEL

After the scenarios were completed they were used to create a shared understanding of the SimSam concept at the HiVe board meeting.

The project leader was asked through the interview if the photorealistic representations had any influence on the decisions made in the school board about SimSam. The project leader argued: “To use

the pictured scenarios was critical for the decisions and has made it easier than understanding it through text, spheres and boxes".

As in any design process these gates of concept evaluation are critical. The evaluation of new concepts at this stage is often problematic because of lack in or proof of data. The project leader expressed that the board was not used to having concepts presented at this early stage with representations that had such a hi-level of fidelity. This made a change in comparison with text, spheres and boxes that can be described as more abstract representation being represented as symbol of things. Using abstract representations as the project leader explains might not reveal this diversity between participants at the decision-making stage. Representations that are pinpointed directly to realistic scenarios with realistic images might help to create a more unified and shared understanding that again can make decision-making easier and more precise.

The illustrated scenarios, created together with the users, made it easier for the board to understand the concept. Because the project had been able to involve expert users so early in the process it gave more reliable data for the board to evaluate.

LEVEL THREE: EXTERNAL COMPANIES AND ORGANIZATIONS

The research results show that partners that were attached to the MarKIS network were able to use the visualizations of SimSam to relate their own practice and areas of interest as possible new SimSam scenarios. When the MarKis network leader was interviewed she stated: "By using the SimSam concept we found possible new ways of changing the existing process of this type of development".

One of the problems she described as part of this type of maritime network is to manage the logistics between partners and interests that can evolve in to new innovation projects. Presenting realistic representations of possible innovation scenarios gave the MarKIS network new frameworks for how to solve these problems.

A critical factor of the MarKIS investment in the project was not only that the SimSam project represented a solution to their maritime innovation network, but also that the realistic representations made it possible to apply their own real-life scenarios into the SimSam concept.

These extracts from the interviews show that through using the photorealistic representations the participation was extended into other research and development networks, where it got applied to new fields. New scenarios created by the external partners have now become central design projects that are being tested in the SimSam-lab.

DISCUSSION AND CONCLUSION

This project has moved forward on two levels simultaneously; (i) through developing the notion of a theoretical framework when working in a multidisciplinary team of both experienced and less experienced participants and (ii) to actually implement these frameworks in the design being developed. The result of the study show how 3D visualizations can be used at different organizational levels to both develop and implement ideas created in the FFE of PD.

There is, however, a paradox concerning the use of hi-fidelity 3D representations in the FFE of a collective design process. The realistic imagery might give the participants an illusion that the design is finished – which will limit the creativity of the design space. On the other hand; using representations that are abstract may result in the participants' lack of understanding the representations. A possible solution for this problem is probably not an optimization of the different factors, but a flexible representation medium.

The layered structure of the 3D model enabled the designers to use the same representation in different stages of the development process. In one stage the representation was used to describe the SimSam idea and technology. On another stage the same representation was used to develop scenarios together with the participants. The developed scenarios were then used to implement the project on a company level and on a level with external

companies and organizations. But could we have achieved the same results using only abstract drawings or physical models, which is a more traditional approach at this stage of the process?

The results show that there was little attention drawn to the project from colleagues when the first abstract sketches of the SimSam idea were made. Some of the participants did not remember that we used the sketches at all. The interviews show that a possible reason for the low initial attention was the low fidelity of abstract drawings and that participants had problems reading them and at the same time relating the drawing to the proposed technology. When creating the scenarios the visual representations needed to be from a bird`s view - capturing all the activity on the lab screens and at the same time being able to zoom in on specific interactions between personas. We also needed to be able to shift between different scenario settings. Fitting these specifications in a physical scale model was not possible (we actually tried this by building a 1:15 scale model using rapid prototyping).

Overall, the case study shows that employment of photorealistic images created by computer generated 3D models can have a useful impact on engaging users and actors as co-designers in the fuzzy front end of product development.

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